Matthew Lau

Cupertino, CA | (408) 596-1554 | https://matthew-coding-portfolio.vercel.app/ | mlau191@cs.washington.edu | https://github.com/mattycakes14

EDUCATION

University of Washington **Seattle, WA**
Bachelor of Science in Computer Science, Paul G. Allen School Graduation: Jun 2028
GPA: 3.71 /4.0 | Dean's List
Coursework: Object-Oriented Programming, Data Structures and Parallelism, Hardware/Software Interface, Linear Algebra, Multivariable Calculus, Probability & Computational Complexity, Machine Learning, Systems Programming

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML, CSS, SQL, Go, C, C++
Frameworks & Libraries: React, React Native, Next.js, Flask, FastAPI, Node.js, Express.js, Tailwind, LangChain, NumPy, TensorFlow, PyTorch**,** Spring
Databases & Tools: DynamoDB, Firebase**,** Supabase, Redis, Qdrant, Pinecone, Docker, Gunicorn, SQLAlchemy, Alembic, AWS (EC2, Auto Scaling Groups, EKS, Lambdas, CloudWatch, Kafka, SNS), Kubernetes, Terraform, Git, GitHub Actions

EXPERIENCE

Amazon **Palo Alto, CA**
Software Engineer Intern – Distributed Databases (Aurora) Sep 2026 – Dec 2026

- Eliminated inaccurate writer-downtime metrics for Aurora Global Database switchovers, caused by start/end events from separate control-plane hosts being mispaired by timestamp proximity due to no correlation ID; designed a per-cluster DynamoDB record to pair events exactly across 3 Java/Spring services, using worker pools to keep DynamoDB reads/writes off the main event-handling thread
- Evaluated 3 designs using a 100-switchover pre-production benchmark and Python tooling; found the simplest fix raced with switchover completion (~75% of pairings missed) and chose a race-free single-record design

Workday **Pleasanton, CA**
Software Engineer Intern – Platform/DevOps Jun 2026 – Sep 2026

- Diagnosed why a fleet-wide OpenTelemetry disable config was silently ignored, found 20 of 34 services still actively exporting, and guarded the export path – eliminating **~632k daily error log spam** and unlocking decommission of the deprecated collector infrastructure (ALB, target groups, ASGs)
- Migrated **60 AWS Lambda functions** from x86-64 to arm64 (Graviton), auditing Docker image layers to verify every binary and transitive dependency resolved to arm64 before cutover; **~20% lower compute cost per GB-second**
- Closed a gap where the monitoring stack (Grafana and Prometheus) was itself unmonitored, provisioning CloudWatch alarms in Terraform on ALB target-group health and routing them through SNS to PagerDuty under escalation policy
- **Top 5 Finalist** among 36 intern teams across the US, Canada, and Costa Rica in Workday's intern hackathon; designed an agentic AI career pilot for early-career talent and pitched it to the VP of Software Development and 100+ attendees

UW Medicine **Seattle, WA**
Software Engineer – Full Stack Feb 2026 – Present

- Rewrote **Firestore** security rules and built Next.js API routes for role assignment, enforcing role-specific database permissions for admins, leads, and volunteers across **100+ service programs**
- Restructured event scheduling data from a nested map field into a per-slot subcollection so security rules could validate writes at document granularity, bounding volunteer signup updates to ±1 within capacity
- Built Firestore test cases on the Firebase emulator with fabricated admin/lead/volunteer auth contexts, and containerized it with Docker so the suite runs identically across machines without a separate Firebase dev project or local JVM

Clad Labs (YC F25) **San Francisco, CA**
Software Engineer Intern – AI Tools Jun 2025 – Sep 2025

- Architected an ad-content extraction pipeline (scraping + OCR) that increased ad-text coverage by **22%** and website-text by **33%,** generating semantic embeddings in Pinecone to match user prompts to relevant ads
- Built Flask API routes to recommend developer-tool advertisements inside AI coding assistant chats, using LangGraph for routing prompts and Supabase for storing user tool preferences for **10,000+ developer waitlist**